

THE 10 CENT WIRE: CUT LENGTHS AND THE FOUR MINUTE BUILD

Your handheld's rubber antenna is only half an antenna. One wire hands it the other half.

A handheld antenna wants a sheet of metal under it, a ground plane, and the radio does not have one. So your hand, wrist, and head fill in, and turn part of every watt into heat. A wire of the right length hung off the antenna nut is the missing bottom half. The return current stops cooking your palm and the signal drops toward the horizon, where the repeater is.

CUT LENGTHS

19 1/2 in	Two meter band, where most of these radios live.
6 1/2 in	The higher band most of these radios also cover.
about 6 in	Family radio channels around 462. Only on a radio with a screw on antenna.

THE FOUR MINUTE BUILD

- 1 Thin insulated wire, any doorbell gauge, and one small ring terminal.
- 2 Cut the wire to length for your band.
- 3 Strip half an inch, then crimp or solder the ring on.
- 4 Unscrew the antenna and drop the ring over the connector threads.
- 5 Screw the antenna back down so the ring is pinched against the connector's metal base.
- 6 Let it hang free. Do not wrap it, hold it, or trim it to look tidy. The length is the point.

FREE RANGE

- ☐ Hold the radio straight up. A repeater antenna stands vertical.
- ☐ Get it away from your body: arm extended, or up on a windowsill.
- ☐ Climb when it is safe. A porch roof 20 feet up pushes your horizon past 6 miles instead of three and a half.
- ☐ Upgrade: a 15 inch aftermarket whip, 15 to 20 dollars, plus the tail underneath, makes a real dipole in your hand.



Radio straight up, arm out, away from your body.

THE HONEST NUMBERS

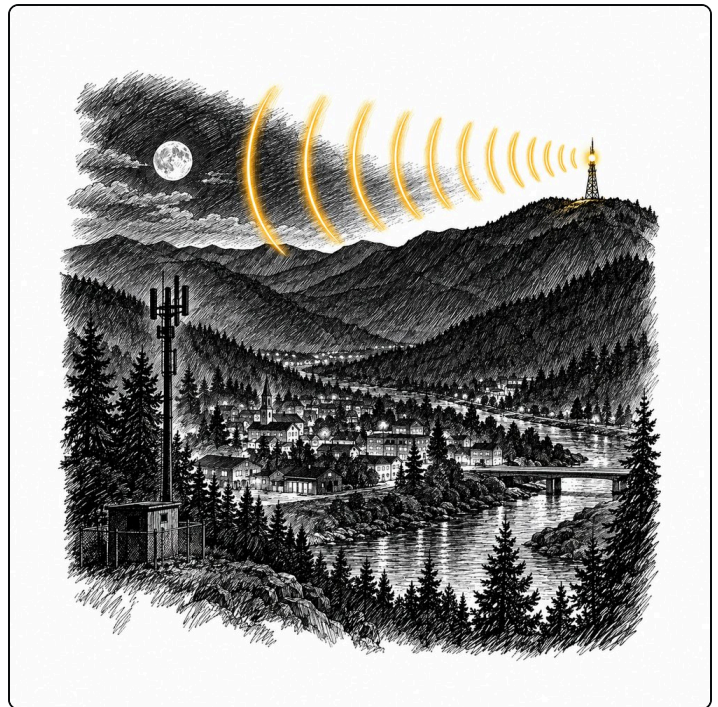
1/5 watt	What a phone transmits. A 25 dollar handheld is rated 5 watts, 25 times that, and needs no tower.
3 dB	Double the power. Hams who measured a tiger tail report a few decibels, depending on radio, band, and grip.
40 percent	What doubling power adds to distance in open air. But range is a cliff: at the edge, a few decibels is unreadable versus readable.
Horizon	Square root of antenna height in feet, times about 1.4, is your radio horizon in miles. 6 feet sees about 3 1/2 miles. A repeater 500 feet up sees about 32.

THE LICENSE, THE LEGAL BUTTON

- ☐ **Amateur Technician:** 35 multiple choice questions from a published pool, no Morse code since 2007, test about 15 dollars, license fee 35 dollars for ten years. It lets you build your own antennas.
- ☐ **General Mobile Radio Service license:** 35 dollars, no exam, ten years, covers your immediate family.
- ☐ License free bubble pack radios have fixed antennas, top out at 2 watts, and cannot work a repeater. No nut, no tail.



Ring terminal over the threads, antenna screwed back down on top of it.



The repeater on the ridge can hear you when the tower in town is dark.

DO THIS ON A CALM AFTERNOON

- ☐ Find your local repeater in an online directory. Check if it is open to all licensed users, and whether it has backup power.
- ☐ Program the channel and the access tone, clip on the tail, and make one test call.
- ☐ Listen before you key. Give your call sign at the end of the exchange and at least every ten minutes.
- ☐ Emergency rule: when human life is in immediate danger and normal systems are down, a station may use any means it has.
- ☐ Cell site batteries generally hold four to eight hours. Satellite texting on newer phones is worth having too, but it wants open sky. Carry both.

Watch the full video: youtube.com/@SketchySurvival101

More free sheets: sketchysurvival101.com/free

Subscribe so the next dead tower does not catch you flat footed.

Sketchy Survival 101. Physics first survival. Every number on this sheet comes straight from the video. Prices move, so check the current listing. Transmitting needs the right license, except in a genuine life threatening emergency.